

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

Reconstruct how the encrypted message might appear if a shift of 4 positions is applied. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

ANSWER:

Scenario

A student organization wants to securely share confidential event details with its core members using a simple classical encryption technique. The original message is:

Plaintext: MEET AT NOON

A Caesar Cipher with a shift of 4 positions is used for encryption.

Encryption Process (Shift = 4)

Each letter is shifted 4 positions forward in the alphabet.

Plain Letter	Encrypted Letter
M	Q
E	I
E	I
T	X
A	E
T	X
N	R
O	S
O	S
N	R

Spaces remain unchanged.

Encrypted Message (Ciphertext)

QIIX EX RSSR

Decryption Process

An authorized member who knows the encryption rule (shift = 4) decrypts the message by shifting each encrypted letter 4 positions backward.

Cipher Letter	Original Letter
Q	M
I	E
I	E
X	T
E	A
X	T
R	N
S	O
S	O
R	N

Decrypted Message

MEET AT NOON

Result

- Plaintext: MEET AT NOON
- Encryption Method: Caesar Cipher
- Shift Value: 4
- Ciphertext: QIIX EX RSSR
- Decrypted Text: MEET AT NOON

Conclusion

The Caesar Cipher encrypts the message by shifting each letter 4 positions forward. An authorized receiver who knows the shift value decrypts the message by shifting each letter 4 positions backward, successfully recovering the original message. This demonstrates a simple classical encryption technique for secure communication.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption. Deduce the most likely original message. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

ANSWER:

Scenario

A cybersecurity intern intercepts the coded message:

Ciphertext: GSRH RH Z HVXIVG

The message appears to use a simple substitution cipher rather than a complex encryption method.

Identifying the Cipher

The given ciphertext is a well-known example of the Atbash Cipher, a monoalphabetic substitution cipher.

In the Atbash Cipher, each letter is replaced by its opposite letter in the alphabet:

Plain Letter	Cipher Letter
A ↔ Z	B ↔ Y
C ↔ X	D ↔ W
E ↔ V	F ↔ U
G ↔ T	H ↔ S
I ↔ R	J ↔ Q
K ↔ P	L ↔ O
M ↔ N	

Decryption Process:

Cipher Letter	Plain Letter
G	T
S	H
R	I
H	S

Cipher Letter	Plain Letter
R	I
H	S
Z	A
H	S
V	E
X	C
I	R
V	E
G	T

Original Message (Plaintext)

THIS IS A SECRET

Justification

The intercepted message matches the characteristics of the Atbash Cipher, where each alphabet letter is substituted with its reverse counterpart ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, etc.).

- The repeated ciphertext pattern "RH" decrypts to "IS", indicating repeated words.
- The single-letter word "Z" decrypts to "A", which is a common English word.
- The final word "HVVXIVG" decrypts to "SECRET", producing a meaningful English sentence.
- The complete decrypted message "THIS IS A SECRET" is grammatically correct and fits the context of a confidential communication.

Result

- Ciphertext: GSRH RH Z HVXIVG
- Cipher Used: Atbash Cipher (Simple Substitution Cipher)
- Decrypted Plaintext: THIS IS A SECRET

Conclusion

The intercepted message was encrypted using the Atbash Cipher, a simple substitution cipher that reverses the alphabet. By replacing each letter with its opposite counterpart, the original message is successfully recovered as: **THIS IS A SECRET**

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality. Demonstrate how the encryption process transforms the message step-by-step. Explain how the repeating keyword influences each character of the ciphertext.

ANSWER:

Scenario

A company uses a keyword-based encryption technique (Vigenère Cipher) for secure internal communication.

- Plaintext: HELLO
- Keyword: KEY

Step 1: Repeat the Keyword

Plaintext	H	E	L	L	O
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Keyword	K	E	Y	K	E
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Step 2: Convert Letters to Numerical Values

Using A = 0, B = 1, ..., Z = 25

Letter	H	E	L	L	O
---------------	---	---	---	---	---

Plaintext Value	7	4	11	11	14
------------------------	---	---	----	----	----

Letter	K	E	Y	K	E
---------------	---	---	---	---	---

Keyword Value	10	4	24	10	4
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Step 3: Encrypt Using the Vigenère Formula

Formula:

Ciphertext = (Plaintext + Keyword) mod 26

Plaintext	H	E	L	L	O
------------------	---	---	---	---	---

Plain Value	7	4	11	11	14
--------------------	---	---	----	----	----

Key Value	10	4	24	10	4
------------------	----	---	----	----	---

Sum (mod 26)	17	8	9	21	18
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Cipher Letter	R	I	J	V	S
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Encrypted Message (Ciphertext)

RIJVS

Explanation

The Vigenère Cipher uses a repeating keyword to determine how much each letter of the plaintext is shifted.

- The keyword KEY is repeated as KEYKE to match the length of HELLO.
- Each letter of the keyword provides a different shift value.
- Unlike the Caesar Cipher, the shift is not constant; it changes according to the corresponding keyword letter.
- This produces a stronger encryption because identical plaintext letters can encrypt to different ciphertext letters depending on the keyword.

Result

- Plaintext: HELLO
- Keyword: KEY
- Repeated Keyword: KEYKE
- Encryption Method: Vigenère Cipher
- Ciphertext: RIJVS

Conclusion

The Vigenère Cipher encrypts the plaintext HELLO using the repeating keyword KEY, resulting in the ciphertext RIJVS. The repeating keyword determines the shift for each character, making the encryption more secure than a fixed-shift substitution cipher.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection. Analyze how investigators could detect the presence of hidden data within the image. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

ANSWER:

Scenario

During a digital forensic audit, investigators suspect that a company logo image contains hidden confidential information. Although the image appears normal, it may have been altered using steganography to conceal secret data.

Detection of Hidden Data

Investigators can analyze the image using the following methods:

1. Visual Inspection

- Compare the suspicious image with the original logo.
- Zoom in to identify unusual pixels, color changes, or image distortions.
- Although most steganographic images appear normal, slight differences may sometimes be visible.

2. Statistical Analysis

- Examine the image's pixel values and color distribution.
- Hidden data often changes the Least Significant Bits (LSBs) of pixel values, creating detectable statistical patterns.

3. Steganalysis Tools

- Use forensic software such as StegExpose, StegDetect, or OpenStego to analyze the image.
- These tools can identify hidden messages or detect signs of steganographic modifications.

4. Metadata Examination

- Check the image metadata for unusual information such as unexpected software, modification dates, or suspicious file properties.

Techniques to Strengthen Hidden Communication

1. Encrypt the Secret Data Before Hiding

- Encrypt the confidential message using algorithms such as AES before embedding it into the image.
- Even if the hidden data is extracted, it remains unreadable without the decryption key.

2. Use Advanced Steganography Techniques

- Instead of simple LSB substitution, use transform-domain methods such as Discrete Cosine Transform (DCT) or Discrete Wavelet Transform (DWT).
- These techniques hide data in the image's frequency components, making detection much more difficult.

Result

- Investigation Method: Digital Image Steganalysis
- Detection Techniques:

- Visual inspection
- Statistical analysis
- Steganalysis tools
- Metadata examination

- Security Recommendations:

1. Encrypt the secret data before embedding.

2. Use advanced steganography methods (DCT/DWT) instead of simple LSB embedding.

Conclusion

Investigators can detect hidden information by performing visual inspection, statistical analysis, metadata examination, and using steganalysis tools. To improve the security of hidden communications, confidential data should first be encrypted and then embedded using advanced steganography techniques such as DCT or DWT. These methods significantly reduce the chances of successful detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”. a. Determine the most probable original message by reversing the encryption process. b. Critically evaluate why such a method is vulnerable in modern communication systems. c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

ANSWER:

Scenario:

A company's legacy communication system uses a transposition-based encryption method with three levels (3 rails). The received ciphertext is:

Ciphertext: WECRLTEERDSOEFEAOCAIVDEN

a. Determine the Original Message

The ciphertext is a classic example of the Rail Fence Cipher (3 Rails).

Decryption Process (3 Rails)

The ciphertext is rearranged by reconstructing the zigzag (rail fence) pattern and then reading the characters row by row.

Rail 1: W . . . E . . . C . . . R . . . L . . . T . . . E

Rail 2: . E . R . D . S . O . E . E . F . E . A . O . C .

Rail 3: .. C ... A ... I ... V ... D ... E ... N

Original Message (Plaintext)

WE ARE DISCOVERED FLEE AT ONCE

b. Vulnerability of the Method

The Rail Fence Cipher is considered insecure in modern communication systems for the following reasons:

- It only changes the positions of letters and does not alter the letters themselves.
- Letter frequencies remain unchanged, making the ciphertext vulnerable to frequency analysis.
- The number of possible rail arrangements is small, allowing attackers to use brute-force attacks.
- Modern computers can decrypt such ciphers within seconds.

c. Practical Enhancement

Use Modern Encryption Algorithms (AES)

A practical improvement is to replace the Rail Fence Cipher with the Advanced Encryption Standard (AES).

Advantages of AES:

- Uses strong cryptographic keys (128, 192, or 256 bits).
- Resistant to brute-force and cryptanalytic attacks.
- Provides strong confidentiality and data protection.

Result

- Ciphertext: WECRLTEERDSOEFEAOCAIVDEN
- Encryption Method: Rail Fence Cipher (3 Rails)
- Original Message: WE ARE DISCOVERED FLEE AT ONCE

Conclusion

The intercepted ciphertext was produced using a 3-Rail Fence Cipher. By reversing the rail arrangement, the original message is recovered as "WE ARE DISCOVERED FLEE AT ONCE." Although the Rail Fence Cipher was useful historically, it is not suitable for modern communication because it offers very weak security. Replacing it with a strong encryption algorithm such as AES provides significantly better confidentiality and protection against modern cyber threats.

